

XIAO SHANG, P.Eng, PhD Candidate

+1 (647) 389-9165 | xiao.shang@mail.utoronto.ca | [Personal website](#) | [Google Scholar](#) | [LinkedIn](#) | Toronto, Ontario, Canada

Education

University of Toronto

Toronto, ON, Canada

Doctor of Philosophy - Materials Science and Engineering (GPA 4.0/4.0)

May 2021 – May 2026

McGill University

Montréal, QC, Canada

Master of Science – Mechanical Engineering (GPA 4.0/4.0)

2015 – 2017

South China University of Technology

Guangzhou, China

Bachelor of Engineering – Mechanical Engineering (GPA 3.8/4.0)

2011 - 2015

Research Experience

Lab for extreme mechanics & additive manufacturing, University of Toronto

Toronto, ON, Canada

Graduate Research Assistant (Supervised by Prof. Yu Zou)

Design and additive manufacture of high performing metallic functionally graded materials via microstructure modification

2021 - Present

MSM@Harwell Group, University College London

Didcot, UK

Visiting Scholar (Supervised by Prof. Peter D. Lee)

Investigate magnetic field assisted additive manufacturing in synchrotron X-ray

Dec. 2023,
Dec. 2024,
Jan. - Feb. 2025

Brunner Neutrino Lab, McGill University

Montréal, QC, Canada

Research Engineer (Supervised by Prof. Thomas Brunner)

Provide engineering support to various nuclear physics experiments

2018 - 2021

Architected materials and advanced structures group, McGill University

Montréal, QC, Canada

Graduate Research Assistant (Supervised by Prof. Damiano Pasini)

Durable bistable auxetics made of rigid solids

2015 – 2017

Micro engineering, dynamics and automation lab, University of Calgary

Calgary, Canada

Mitacs Globalink research intern (Supervised by Prof. Simon Park)

Investigate the effects of sapphire tooling on the machining process

June - Sept. 2014

Publications and Awards

Selected publications (Full list at [Google Scholar](#))

1. **Shang, X.**, Dash, S. S., Shao, C., Lang, L., ... & Zou, Y. (2026). Confined necking and improved tensile ductility in heterostructured bi-metallic steels made by additive manufacturing. *Acta Materialia*, 121952. 2026
2. **Shang, X.**, Evelyn L., Soumya S. D., Ajay T., & Zou, Y. (2026). Additive manufacturing of multi-metallic materials to achieve multi-functionality. *MetalMat*, e70032. 2026
3. **Shang, X.**, Talbot, A., Li, E., Wen, H., Lyu, T., Zhang, J., & Zou, Y. (2025). Accurate inverse process optimization framework in laser directed energy deposition. *Additive Manufacturing*, 102, 104736. 2025
4. **Shang, X.**, Liu, Z., Zhang, J., Lyu, T., & Zou, Y. (2023). Tailoring the mechanical properties of 3D microstructures: A deep learning and genetic algorithm inverse optimization framework. *Materials Today*, 70, 71-81. 2023
5. **Shang, X.**, Liu, L., Rafsanjani, A., & Pasini, D. (2018). Durable bistable auxetics made of rigid solids. *Journal of Materials Research*, 33(3), 300-308. 2018
6. **Shang, X.**, Yang, S., Chapman, H., Fan, X., ... & Zou, Y. (2025). Directed energy deposition additive manufacturing: advanced microstructure and composition engineering for high-performing metallic materials. Invited review paper under review at *Additive Manufacturing*. 2026
7. Zhang, J.*, **Shang, X.***, ... & Zou, Y. (2025). Mechanism and quantification of melt pool morphology evolution in single-track fabrication by laser direct energy deposition. *Journal of Manufacturing Processes*, 156, 556-568. *indicates co-first authors. 2025
8. Chen, H., Lang, L., **Shang, X.**, Dash, S. S., He, Y., King, G., & Zou, Y. (2024). Anisotropic co-deformation behavior of nanolamellar structures in additively manufactured eutectic high entropy alloys. *Acta Materialia*, 271, 119885. 2024
9. Talbot, A., **Shang, X.**, ... & Zou, Y. (2025). Laser remelting of a CrMnFeCoNi high entropy alloy: Effect of energy density on elemental segregation. *Advanced Engineering Materials*, 2501194. 2025
10. Yang, S., Goodall, D. D., Jin, X., **Shang, X.**, ... & Chechik, L. (2026) Effect of Wall Thickness on Heat Flow Conditions in Directed Energy Deposition. *International Journal of Mass and Heat Transfer*, 260 (2026): 128414. 2026
11. Dash, S. S., Fernandes, R. C., **Shang, X.**, Zou, Y., ... & Chen, D. (2025). Fatigue and deformation mechanisms of ultrasonic spot-welded dissimilar joints of a magnesium alloy to a clad aluminum alloy. *Journal of Magnesium and Alloys*. 2025
12. Dash, S. S., Lang, L., Agyapong, J., Yang, S., **Shang, X.**, ... & Zou, Y. (2025). Hetero-deformation and fracture of a near-eutectic high entropy alloy under dynamic impact. *Scripta Materialia*, 260, 116593. 2025
13. Wang, W., Choi, H. S., Dash, S. S., Lyu, T., **Shang, X.**, & Zou, Y. (2025). Self-generating lubricious oxides for friction reduction in FeCoNiMo and CrCoNiMo from room temperature to 1000° C. *Journal of Alloys and Compounds*, 1022, 180003. 2025
14. Zhang, J., Jiang, R., Li, K., Chen, P., Bi, S., **Shang, X.**, ... & Zou, Y. (2025). Accurate and efficient predictions of keyhole dynamics in laser materials processing using machine learning-aided simulations. *International Journal of Heat and Mass Transfer*, 250, 127279. 2025
15. Lyu, T., Keshavarz, M. K., Patel, S., Haché, M. J., Cheng, C., **Shang, X.**, ... & Zou, Y. (2024). Melting mode driven tuning of local microstructure and hardness in laser powder bed fusion of 18Ni-300 maraging steel. *Materialia*, 35, 102113. 2024

16. Yin, Z., ... **Shang, X.**, ... (2021). Young martlets: Exploring the world of academia and beyond. *Matter*, 4(5), 1434-1436. 2021

Conference presentations

1. **Shang, X.**, Guo, L., & Zou, Y. (2026). Radial Bi-Metallic Materials to Overcome Strength-Ductility Trade-Off Under High-Speed Impact. *The 155th TMS Conference*. Oral presentation. 2026
2. **Shang, X.*** & Zou, Y. (2026). Confined necking and improved tensile ductility in heterostructured bi-metallic steels made by additive manufacturing. *The 155th TMS Conference*. Poster presentation. 2026
3. **Shang, X.** (2025) Autonomous Additive Manufacturing System. *ASM International IMAT Materials Venture Challenge*. Oral presentation. **First place**. 2025
4. **Shang, X.*** & Zou, Y. (2025). Directed energy deposition of heterostructured steels for optimal mechanical properties. *The 154th TMS Conference* and *HI-AM Conference*. Oral presentation. 2025
5. **Shang, X.*** & Zou, Y. (2025). Accurate inverse process optimization framework in laser directed energy deposition, *The 154th TMS Conference*. Oral presentation. 2024
6. **Shang, X.***, Dash, S. S., Lyu., T., Talbot, A., ... & Zou, Y. (2024). Investigate the mechanical properties of DED manufactured heterogenous steel for magnetic multi-functional materials, *The 153rd TMS Conference*. Oral presentation. 2024
7. **Shang, X.*** & Zou, Y. (2023). An inverse materials design route based on structure-property linkages using 3D convolutional neural network and Bayesian optimization. *The 152nd TMS Conference & Acceleration Consortium Conference*. Oral and poster presentations, **Best poster award**. 2023
8. **Shang, X.***, Liu, X., ... & Zou, Y. (2023). A deep learning enabled reliability enhancement system for the fused deposition modelling process, *The 152nd TMS Conference*. Poster presentation. 2023

Patents

1. “Multi-material tooth-free shaft in magnetic-fluid seal”. Pending US Patent. 2025
2. “Data encoding via metal additive manufacturing”. Pending US Patent. 2025
3. “Autonomous additive manufacturing system”. US patent in preparation. 2026
4. “Bistable auxetics” (US20170362414). US Patent and Trademark Office. 2020
5. “Modular ground-stowable workstation” (ZL2014103573363). Chinese IP office. 2016

Selected Awards

1. DND/NSERC supplemental funding award, national award (\$10,000) 2026
2. ASM IMAT Materials Venture Challenge 1st Place, international award (\$5,000) 2025
3. Joseph R. Benedetto Scholarship, provincial award (\$1,000) 2025
4. MITACS Globalink Research Award, international award (\$6,000) 2024
5. NSERC Michael Smith Foreign Study Supplements, national award (\$6,000) 2024
6. NSERC Canada Graduate Scholarships – Doctoral (CGS D), national award (\$120,000) 2023
7. Ontario Graduate Scholarship, provincial award, twice (\$30,000) 2021 & 2022
8. MITACS Globalink Graduate Fellowship, international award (\$15,000) 2015

9. MITACS Globalink Research Intern Scholarship, international award (\$7,500)	2014
10. Chinese National Encouragement Scholarship, national award (¥5,000, top 2%)	2014
11. Chinese National Scholarship, national award (¥8,000, top 1%)	2013

Teaching

CIV1199: Joining, Welding and Additive Manufacturing <i>University of Toronto</i> Teaching Assistant	<i>Fall 2025</i>
MSE403 – Data Sciences and Analytics for Materials Engineers <i>University of Toronto</i> Teaching Assistant	<i>Winter 2023, Winter 2024</i>
MSE1065 – Application of Artificial Intelligence in Materials Design <i>University of Toronto</i> Teaching Assistant	<i>Winter 2021, Fall 2022, Fall 2023</i>
ME243 - Manufacturing and CAD Modeling <i>Shijiazhuang Tiedao University</i> Guest Lecturer	<i>Spring 2024</i>
Subatomic Physics Summer Lecture Series <i>McGill University</i> Guest Lecturer	<i>2020-2022</i>

Service

SME Toronto Chapter 26

Website Program manager	<i>2025 - Present</i>
Contribute to program development, event coordination, and member engagement. Organize technical seminars, networking events, and outreach initiatives that connect academia and industry.	

The Minerals, Metals & Materials Society (TMS)

Website Member of the Solidification Committee	<i>2025 - Present</i>
Engage in the program development of the committee. Help shape the conference agenda, suggest panels and special topics. Contribute to long-term planning of the committee's role in the conference.	

Materials Science and Engineering Graduate Students' Association

<i>University of Toronto</i> President	<i>2021– 2022</i>
Responsible for the direction and actions of the association and represent our graduate students at the department, faculty, and university level councils.	

The Martlets Society

Website Organizer and Editor-in-chief	<i>2020 - Present</i>
The Martlets Society is a non-profit organization funded and run by early career researchers. We invite researchers from top universities to give talks on their research topics and career advice.	